

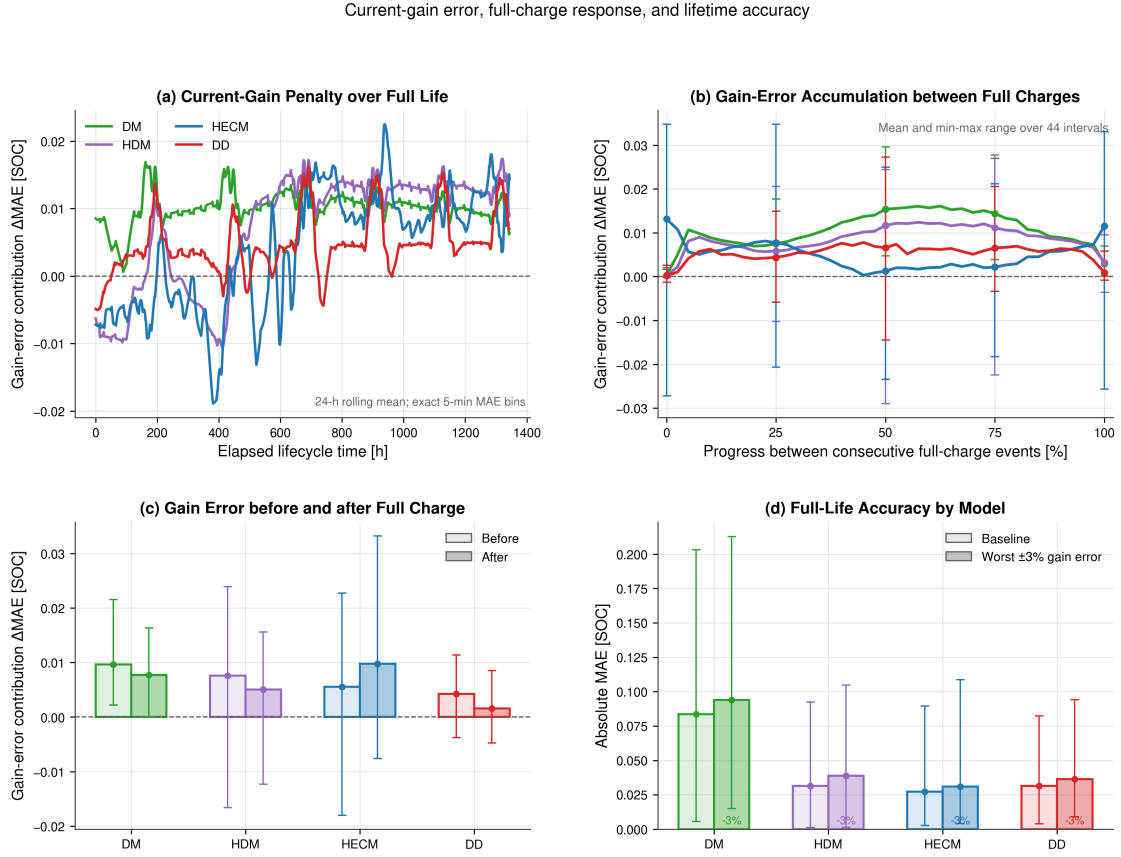


Figure 6.6: Current-gain diagnostic over the continuous high-load holdout trajectory. The panels report the rolling 24-h gain-error contribution over elapsed life, its normalized development between consecutive full-charge events, its value immediately before and after those events, and the full-life baseline and adverse-gain MAE. Full-charge anchoring changes accumulated error but does not eliminate the lifetime penalty uniformly.

6.4.3 Additive sensor offsets

The signed current-offset experiment shifts measured current by ± 0.05 A and retains the larger paired Δ MAE within each cell before equal-weight aggregation. DD has the smallest adverse cell-macro penalty at 0.0401 with a 95% confidence interval from 0.0226 to 0.0556. HECM follows at 0.1763 from 0.1329 to 0.2142, DM at 0.2467 from 0.2253 to 0.2693, and HDM at 0.2727 from 0.2546 to 0.2911. The adverse sign differs between models and cells, showing why one fixed sign would not represent offset sensitivity reliably.

The integration-based estimators accumulate the persistent zero-current error